



ENT-164 · Intro to Making

Tufts
UNIVERSITY

Derby Entrepreneurship
Center at Tufts

CLASS 5 · WEEK 5 · FALL 2026

Electronics

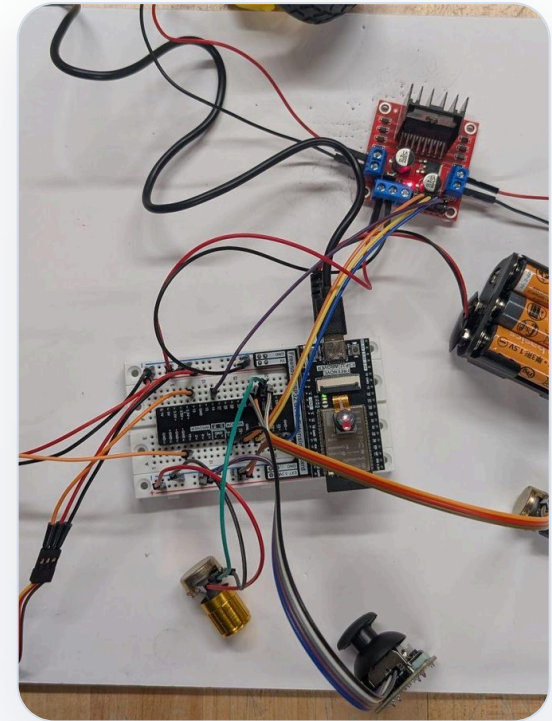
Today your robot gets a brain. We wire motors, sensors, and controls on the breadboard, install MicroPython, and make AI our coding partner.

Thu, Oct 8

JCC 301 / Nolop

Robo Challenge

Bring your ESP32 kit



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Tufts University · Gordon Institute

Seven weeks from sketch to race day

Today is Week 5. Basic Electronics and Microcontrollers. Here's how the robot build comes together.

Week 2

Ideation and Prototyping

Sketch concept variations; build a foam core mockup.

Week 3

From Physical to Digital: 2D

Create a laser-ready 2D design; cut structural parts.

Week 4

From Physical to Digital: 3D

Model 3D parts; 3D print functional parts.

Week 5

Basic Electronics and Microcontrollers

Wire motors, sensors, and controls; program with MicroPython + AI.

Week 6

AI Coding Agents

Connect to a wireless network; build a remote control interface.

Week 7

Final Robot Assembly and Testing

Final integration and test runs.

Week 8

Robo Challenge and Final Project Kickoff

Demonstrate the robot with mechanical systems and motors mounted.

TODAY

Week 5 · Electronics. Wire motors, sensors, and controls on a breadboard, and program basic movements using MicroPython and AI.

Today the robot gets a brain

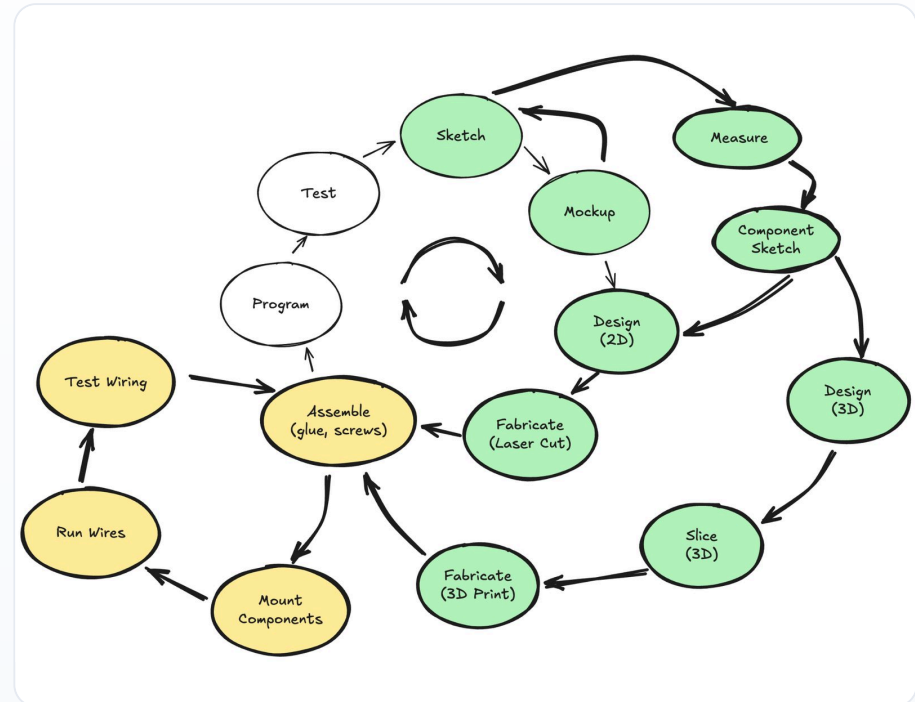
The mechanics are nearly done. Today we bring the electronics to life.

TODAY'S PLAN

- Meet your kit and the ESP32 board
- Install the MicroPython operating system
- Blink an LED with opencode
- Move a servo with AI-generated code
- Wire the robot at Nolop

WHERE WE ARE IN THE BUILD

Sketch, mockup, design, and fabrication are behind you. Now we assemble, run wires, test the wiring, program, and test again.



What's in your **personal kit**?

Freenove FNK0046 Super Starter Kit for ESP32 WROVER. Everything you need for the electronics project.

The brain

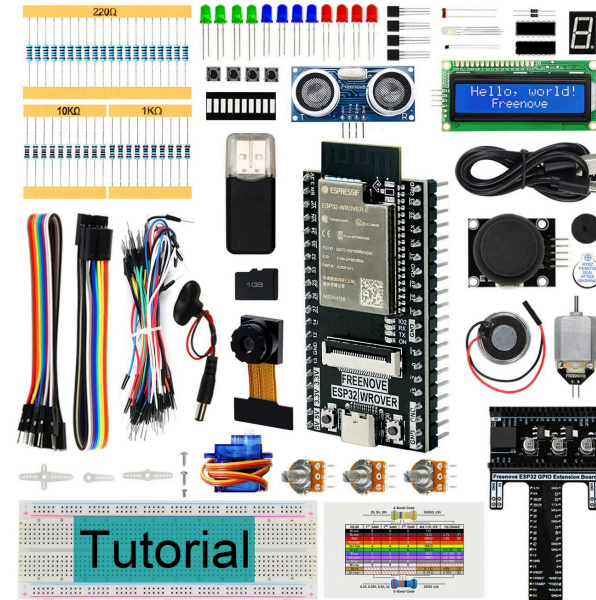
ESP32-WROVER board, GPIO extension board, and a solderless breadboard.

The basics

Jumper wires, LEDs, resistors (220 Ω , 1K, 10K), push buttons, USB cable.

The fun stuff

LCD and 7-segment displays, buzzers, joystick, potentiometers, servo, TT motor + wheel, motor driver, ultrasonic sensor, camera.



NOTE

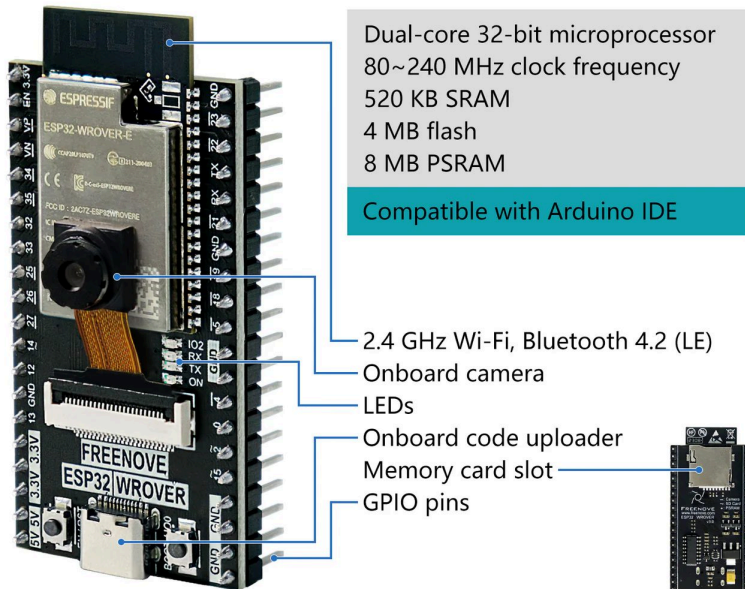
No paper tutorial. The download link for the tutorial and code is **on the box**.

Meet the ESP32-WROVER

A powerful, small controller with onboard wireless and camera.

ESP32-WROVER Board

A powerful, generic and small-sized controller with onboard wireless and camera.



SPECS

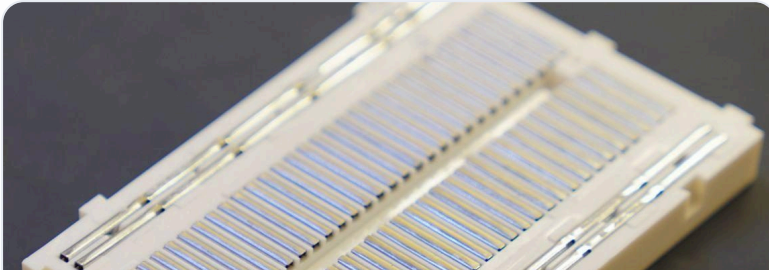
- Dual-core 32-bit microprocessor, 80–240 MHz
- 520 KB SRAM · 8 MB PSRAM · 4 MB flash
- 2.4 GHz Wi-Fi and Bluetooth 4.2 LE
- Onboard camera, LEDs, and code uploader
- Memory card slot and GPIO pins

WHAT IT CONTROLS

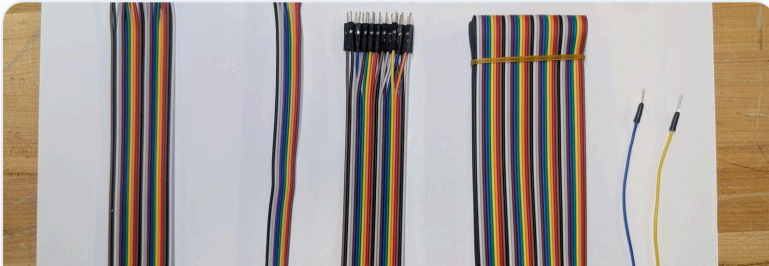
Motors through the driver, servos, joystick, potentiometers, sensors, and displays. The brain of your robot.

How the **breadboard** works

No soldering. Push parts and wires into the holes and they connect.



A solderless breadboard. Rows of connected holes



Jumper wires. Male–male, male–female, female–female

Rows

Each row of five holes is connected horizontally, and the center channel separates the two sides.

Power rails

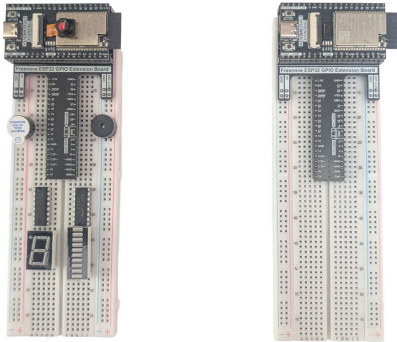
The vertical rails along the edges carry 5V, 3.3V, and GND to every row.

Pin rows

Your ESP32 plugs into the GPIO extension board: any of the horizontal holes in a pin's row connects to that pin. For example, pin 34.

Prep your board

Three steps before the first circuit.



1 · Remove the pre-installed demo parts

02

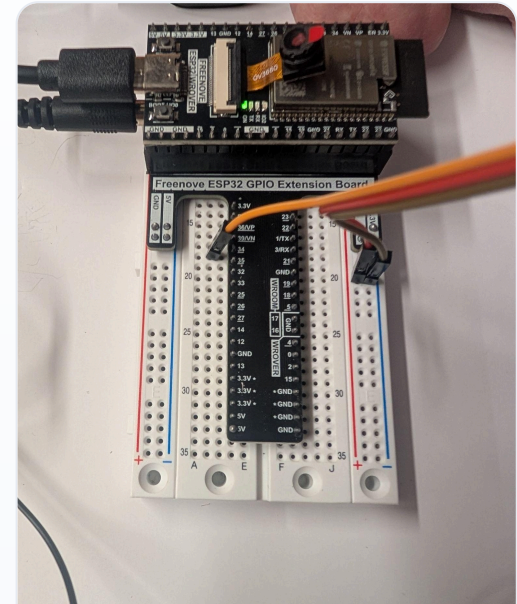
Seat the board

Clip the ESP32 and its GPIO extension board across the breadboard's center channel.

03

Read the labels

5V, 3V3, GND, and the numbered GPIO pins are printed on the extension board. Check before you plug anything in.



Pin labels on the extension board

Installing the **operating system**

Your ESP32 runs MicroPython. A tiny version of Python.

ASK OPENCODE

"Put the snake on the plane". Opencode flashes MicroPython onto your ESP32.

One-time setup

Do this once per board. After that, opencode runs your code on the robot.



The MicroPython snake needs a plane. Your ESP32 board.

Make an LED blink

Ask opencode to write it, then run it on the board.

BLINK.PY

```
from machine import Pin
import time

led = Pin(2, Pin.OUT)

while True:
    led.on()      # Turn LED on
    time.sleep(1) # Wait 1 second
    led.off()     # Turn LED off
    time.sleep(1) # Wait 1 second
```

RUN IT

Ask opencode: “Run this blink program on my ESP32.”

It puts the code on the board and shows you what the program prints.

CHALLENGE

Can you make it blink faster or slower?

Generate code and wiring with AI

Go to claude.ai and enter this prompt.

THE PROMPT

I have an ESP32 WROVER with MicroPython and an SG90 servo.

I want to move the servo arm from 0 to 180 degrees and back.

Generate the code and provide beginner-friendly wiring instructions.

✶ Hey there, Maker

I have an ESP32 WROVER with MicroPython and an SG90 servo.

I want to move the servo arm from 0 to 180 degrees and back.

Generate the code and provide beginner-friendly wiring instructions.

+

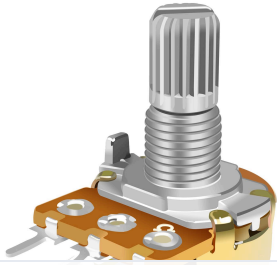
Opus 4.6 ▾



Describe your parts, your goal, and ask for beginner-friendly wiring. One sentence per idea.

Knobs, sticks, and sensors

Your robot reads the world through inputs. Here's how two common ones wire up.



Potentiometer. A knob the ESP32 can read

POTENTIOMETER

- Left pin → GND
- Middle pin (wiper) → ESP32
- Right pin → 3V3



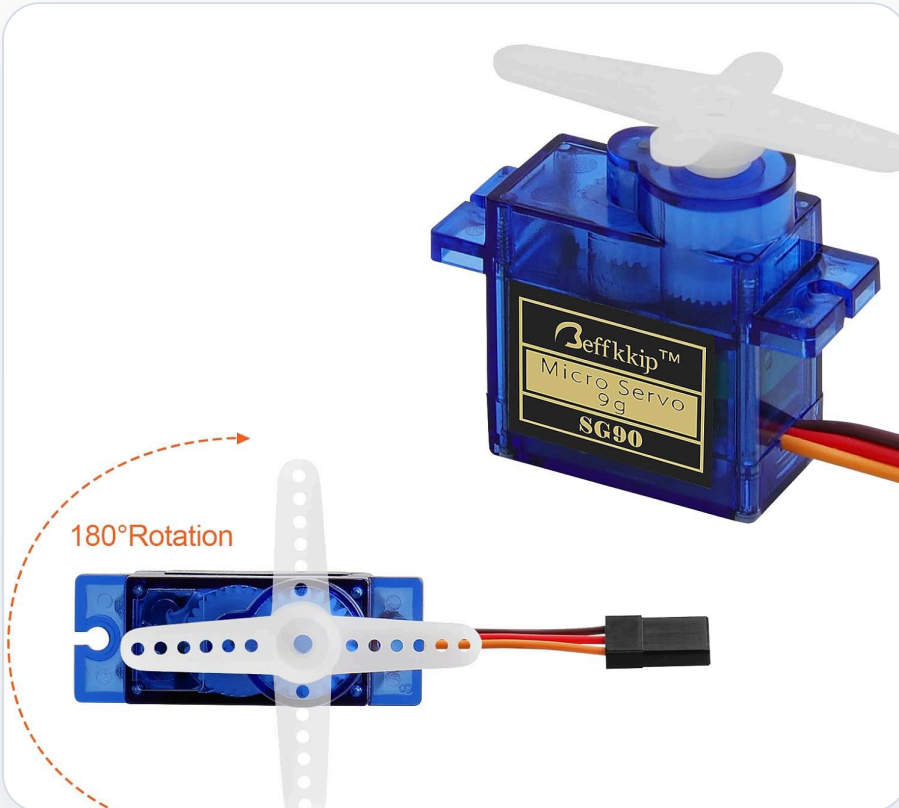
KY-023 joystick. Two axes plus a button

KY-023 JOYSTICK

- GND → GND
- +5V → 3V3
- VRx (X-axis) → ESP32
- VRy (Y-axis) → ESP32
- SW (button) → ESP32

Move an arm with a servo

The SG90 micro servo rotates about 180°. Perfect for arms, gates, and grippers.



SERVO WIRING

- Brown / Black → GND
- Red → 5V
- Orange / Yellow → ESP32

ON THE ROBOT

Servo 1 and Servo 2 wire up the same way. Match the colors, not the memory.

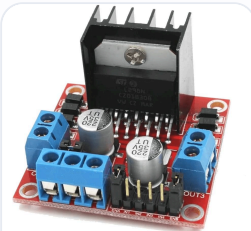
Spin the wheels with the L298N

The driver is the muscle: the ESP32 sends signals, the L298N sends power to the motors.



TT Motor

DC gearbox
motor + wheel



L298N driver

Dual H-bridge
motor driver

L298N WIRING

- 12V → battery pack
- 5V → logic power
- GND → common ground
- IN1, IN2 → ESP32 (left motor)
- IN3, IN4 → ESP32 (right motor)
- OUT1, OUT2 → left motor (red, black)
- OUT3, OUT4 → right motor (red, black)

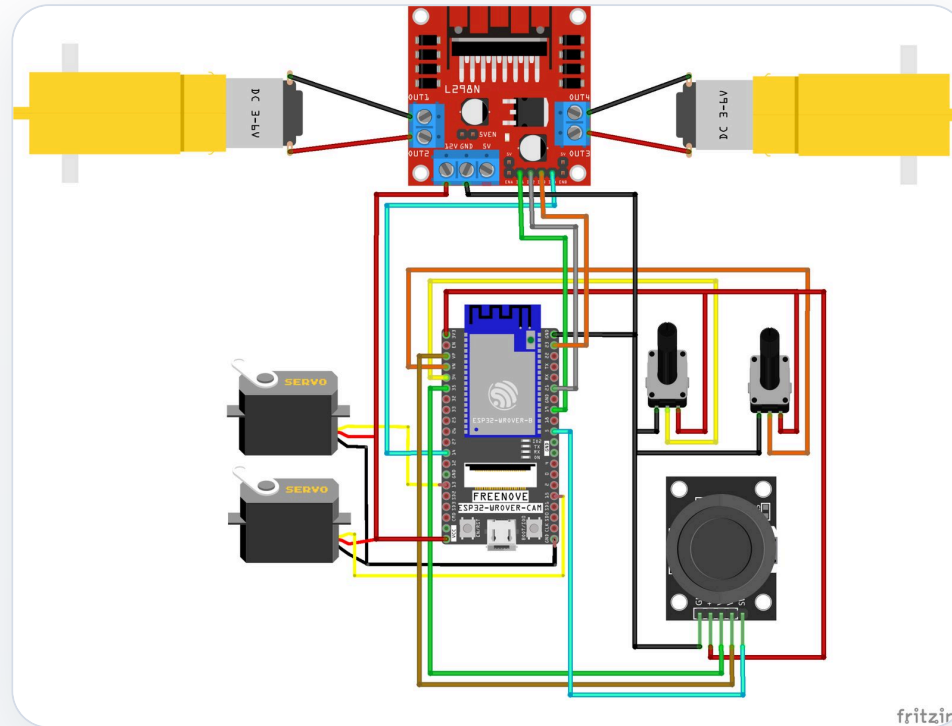
Your robot's wiring map

Two servos, two DC motors, two joysticks, one driver, one brain.

READ IT LIKE A MAP

- Power comes from the battery pack
- Motors run through the L298N driver
- Servos and controls go to GPIO pins

Wire from a diagram, never from memory, and ask AI to explain any wire you don't recognize.



Wire your robot **one part at a time**

Mechanical assembly review, then AI-assisted wiring at the makerspace.

TODAY AT NOLOP

- Mechanical assembly review
- Use AI to wire robot components one by one
- Mount parts and run wires after functional testing

TIPS

- Do not generate all the code and wiring in one prompt.
- Add components step by step. Test and verify with each iteration.
- Provide detailed model numbers for the parts.

NAME YOUR PARTS EXACTLY

- DC Gearbox Motor. "TT Motor"
- BOJACK L298N Motor DC Dual H-Bridge Motor Driver
- KY-023 Control Stick
- SG90 9g Micro Servo Motor
- Freenove ESP32 WROOM

AI-assisted electronics project

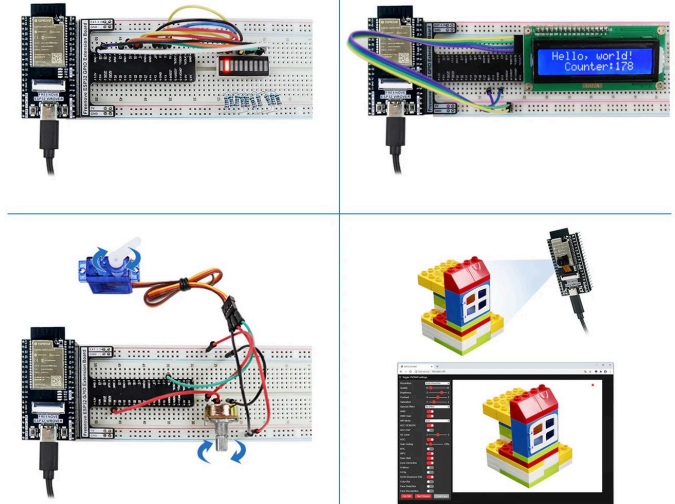
Design and build a fun electronics project using your ESP32 kit. With AI as your coding partner.

REQUIREMENTS

- At least 1 input (sensor, button, etc.) + 1 output (LED, display, buzzer, etc.)
- AI-generated MicroPython code, tested and refined

THE PROCESS. ALL IN ONE CHAT

- **Brainstorm:** attach the parts list image, ask AI for project ideas
- **Build:** get wiring instructions and code from AI
- **Test & iterate:** describe problems, share errors, refine with AI's help



Need ideas? The kit tutorial has example projects and code to borrow from.

What you'll hand in

01

Your AI conversation

A link to your chat. One continuous conversation, start to finish.

02

Video demo

A demo of the working project with a voice-over explanation.

03

Written reflection

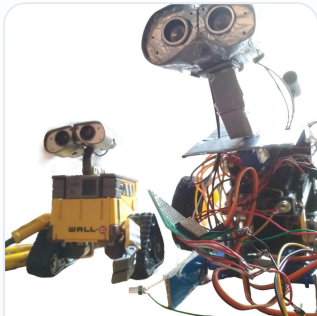
A written reflection on the process.

DEMO Bring your electronics project to class for a demo.

GRADING You're graded on **collaboration and learning**, not coding skills. Keep one chat, iterate often, and ask lots of questions.

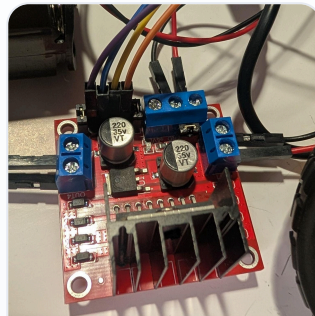
Robots under construction

Wiring, testing, and mounting. The messy middle of the build.



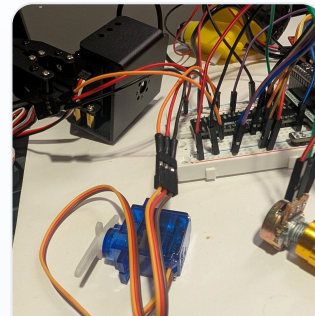
Wire by wire

The robot takes shape.



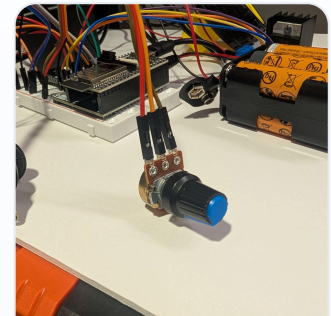
Motor driver

Powered and wired in place.



Test rig

Test each part before mounting.



Controls

Knobs wired and verified.

BEFORE YOU GO

Questions?

Next week: AI Coding Agents. Connect your microcontroller to a wireless network and build a remote control interface.

WHERE TO FIND US

Office hours

go.tufts.edu/meet_tvdv

Email

thomas.van_de_velde@tufts.edu

Course materials

tuftsmaker.github.io/ENT-164